CryptoLens

Phase 2 Data Dictionary (Revised)

Covers every table written by 01.2_bronze_ingest and 02_silver_transform. All schemas are defined explicitly via StructType/StructField in each notebook — no inferSchema is used anywhere in the pipeline.

Revision note: volume, quote_asset_volume, taker_buy_base_vol, and taker_buy_quote_vol in silver_klines were widened from DECIMAL(18,8) to DECIMAL(28,8) after real top-50 data revealed that low-priced, high-supply symbols can trade volumes exceeding DECIMAL(18,8)'s 10-digit integer-part limit. Price columns (open/high/low/close) remain DECIMAL(18,8), which is sufficient for all tracked symbols.

Bronze Layer

bronze_klines

Raw kline (candlestick) records, landed close to source with audit metadata attached. Upserted via MERGE on the natural key below — immutable once written except for the rare case of a new file for the same key.

Column	Type	Notes
open_time	LONG	Epoch ms. Part of natural key.
open, high, low, close	DOUBLE	As published by Binance
volume	DOUBLE	Base-asset volume
close_time	LONG	Epoch ms
quote_asset_volume	DOUBLE	
number_of_trades	LONG	
taker_buy_base_asset_volume	DOUBLE	
taker_buy_quote_asset_volume	DOUBLE	
ignore	STRING	Unused Binance field, kept for fidelity
symbol	STRING	Part of natural key. e.g. BTCUSDT
interval	STRING	Part of natural key. e.g. 1m
load_type	STRING	Part of natural key. full or incremental
source_file	STRING	Volume path of the file this row came from
ingestion_date	DATE	Date the pipeline ran (not business date)
load_timestamp	TIMESTAMP	Exact ingestion instant

Natural key: (symbol, interval, load_type, open_time). Idempotency: MERGE … WHEN NOT MATCHED THEN INSERT — re-running on the same file inserts 0 new rows.

bronze_exchange_info

One row per pipeline run, holding the full raw exchangeInfo JSON response as text. Append-only (each row is a distinct point-in-time snapshot).

Column	Type	Notes
raw_json	STRING	Full raw API response, unparsed
source_file	STRING	Volume path of the landed JSON file
load_timestamp	TIMESTAMP	Snapshot instant — Silver picks the latest for dim_symbol

bronze_quarantine

Rows from any kline file that failed to parse against RAW_KLINE_SCHEMA (schema drift / malformed data). Same shape as bronze_klines plus the populated _corrupt_record column holding the raw offending line.

Silver Layer

silver_klines

One conformed fact row per (symbol, interval, open_time). This is the dedicated fact table analytical/Gold queries read from.

Column	Type	Notes
symbol	STRING	Part of natural key
interval	STRING	Part of natural key
open_time	TIMESTAMP	Cast from Bronze epoch ms. Part of natural key.
close_time	TIMESTAMP	Cast from Bronze epoch ms
open, high, low, close	DECIMAL(18,8)	Cast from DOUBLE for financial precision (prices stay within 10 integer digits)
volume, quote_asset_volume	DECIMAL(28,8)	Wider than price columns: high-supply symbols overflow DECIMAL(18,8)
number_of_trades	INTEGER	Cast from LONG
taker_buy_base_vol, taker_buy_quote_vol	DECIMAL(28,8)	Renamed + cast from Bronze's longer field names; wide for same reason as volume
load_timestamp	TIMESTAMP	When this row was processed into Silver

Natural key: (symbol, interval, open_time). Idempotency: MERGE … WHEN MATCHED THEN UPDATE … WHEN NOT MATCHED THEN INSERT — handles both re-runs and the case where a later full-load file corrects a value from an earlier daily incremental file.

Quality rules (quarantine-by-filter): rows dropped if high < low, or if close/open fall outside [low, high].

dim_symbol

One row per trading pair — the one table in this pipeline that demonstrates a true update/soft-delete, since kline data itself is append-only.

Column	Type	Notes	Key
symbol	STRING	e.g. BTCUSDT	PK
base_asset	STRING	e.g. BTC	
quote_asset	STRING	e.g. USDT	
status	STRING	TRADING, BREAK, DELISTED, etc.	
is_spot_trading_allowed	BOOLEAN		
is_active	BOOLEAN	false once soft-deleted	
load_timestamp	TIMESTAMP	Last time this row changed	

- WHEN MATCHED AND status/filters changed → UPDATE
- WHEN NOT MATCHED BY TARGET → INSERT (new listing)
- WHEN NOT MATCHED BY SOURCE → soft-delete (is_active = false, status = 'DELISTED') — history is preserved, never hard-deleted.

Operational / Audit

pipeline_execution_logs

One row per file/batch processed, at every layer, every run.

Column	Type	Notes
run_id	STRING	UUID, unique per log entry
layer	STRING	Raw-to-Bronze or Bronze-to-Silver
parameter_processed	STRING	File path (Bronze) or symbol-interval-loadtype (Silver)
load_type	STRING	full or incremental
start_time	TIMESTAMP	

Column	Type	Notes
end_time	TIMESTAMP	
status	STRING	SUCCESS, FAILURE, or SKIPPED
rows_inserted	LONG	
rows_updated	LONG	
error_message	STRING	NULL on success

No primary key enforced (append-only log); query by run_id or filter on layer/status/time range for monitoring.